Session 21 · Thursday 5 November 2026 · A home guide. Sections 1 and 2 re-run Thursday's derivation so you can do it cold. Sections 3 and 4 go past it — the numbers there are on no slide. Answers are released after the session.

The same derivation, four times

One regulator X, made at rate k and removed at rate γ. Downstream sits a module that reads it: p_{tot} binding sites, dissociation constant K_d. Binding is fast compared with everything else.

$$X + p \rightleftharpoons C, \qquad C(X) = \frac{p_{\text{tot}} X}{K_d + X}$$

Throughout, write $X_T = k/\gamma$ for what the regulator would reach with nothing listening, and use the reference numbers $k = 10$, $\gamma = 0.01$ — so $X_T = 1000$ — with $K_d = 1$ and $p_{\text{tot}} = 300$ when a number is wanted.

One physical question decides everything below, and it is worth writing at the top of your own page before you start: *while the regulator is sitting in the complex, is it still being removed?*

At each transition the sheet asks *why does that step follow?* Answer in writing before you go on.

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1 · Fully worked — where the retroactivity comes from

The total. Free plus parked:

$$X_T^{\text{inst}} = X + C(X)$$

Differentiate. C depends on time only through X, so the chain rule gives

$$\frac{dX_T^{\text{inst}}}{dt} = \left(1 + \frac{dC}{dX}\right)\frac{dX}{dt}$$

Name the derivative. That bracket is the whole of it:

$$R(X) \;=\; \frac{dC}{dX} \;=\; \frac{p_{\text{tot}} K_d}{(K_d + X)^2}$$

So the dynamics. With $f(X)$ the law the **total** obeys,

$$\boxed{\frac{dX}{dt} = \frac{f(X)}{1 + R(X)}}$$

Read it. A load does not add a term. It **divides the rate**, by a factor that depends on where you are. At $X = 0$ the division is by $1 + p_{\text{tot}}/K_d$; once $X \gg K_d$ the sites are full, $R \to 0$, and the division goes away.

▶ **Why does that step follow?** The load never touched f, so it is tempting to conclude that the steady state is untouched too. But nothing above said what f is. Name the one physical fact about the complex that decides it, and say in one sentence why the conclusion needs that fact to go one particular way.

2 · Last step blank — the complex is protected

Suppose the site **shields** the regulator: while bound, it is not degraded and not diluted. Then only free X is removed, so the total obeys

$$\frac{d(X+C)}{dt} \;=\; k - \gamma X$$

▶ **Setting the left side to zero gives $X_{ss} =$ _____, which depends on p_{tot} how?** _____

▶ **Why does that step follow?** You have just shown there is no steady-state cost at all. But the response time *does* change. In one sentence: what is the load costing you, if not level?

3 · More blank — the complex is diluted

Now suppose the complex is removed like everything else — the cell is growing and dilutes bound and free alike. Then **both** terms are removed:

$$\frac{d(X+C)}{dt} \;=\; k - \gamma X - \gamma C$$

(a) Set the left side to zero. What is pinned now?

$$X + C(X) \;=\; _____$$

(b) Use $K_d = 500$ for the rest of this section, not the $K_d = 1$ at the top of the sheet. The sites are no longer anywhere near full, so you cannot write $C \approx p_{\text{tot}}$ and you have to solve it. Substitute $C(X)$ into your answer to (a) and clear the denominator. You should get a quadratic:

$$X^2 + (K_d + p_{\text{tot}} - X_T)\,X - X_T K_d \;=\; 0$$

(c) Put the numbers in — $p_{\text{tot}} = 300$, $X_T = 1000$, $K_d = 500$ — and take the positive root.

$$X_{ss} = _____ \qquad \frac{X_T - X_{ss}}{X_T} = _____\%$$

(d) The saturated shortcut — assume the sites are full, so the drop is just p_{tot}/X_T — would have answered 30%. Your answer is smaller. Compute $C(X_{ss})$ and you will find it is exactly the number of molecules free X lost. Say in one sentence what sets the size of the drop, given that it is not the site count.

▶ **Why does that step follow?** You now have two models that disagree about whether the level moves, and the algebra cannot choose between them. Design the **measurement** that does. One sentence, and say what each outcome would mean.

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4 · Entirely yours — what affinity does and does not touch

(a) Compute $\int_0^\infty R(X)\,dX$. You do not need to integrate anything — you need to notice what R is the derivative of.

(b) Say in one sentence what that integral *means* physically, in terms of molecules the loaded system has to make on its way up.

(c) Hence: a colleague proposes escaping the load by engineering a weaker operator — raising K_d by two orders of magnitude. What happens to the response time, and why? Then say what *does* change, and what it costs them.

(d) Thursday's board got the protected model's weak-load slowdown as a clean multiple of p_{tot}/X_T, by counting the extra molecules the loaded system must make before it reaches *the same* half-way point. **That argument fails in the diluted model.** Say which step of it breaks, and why.

(e) Thursday's figure showed the diluted ratio running from about 1.44 at the light-load end to about 1.71 at the heavy end. Using your answer to (d), say why **no** constant can be right there — not a different constant, none.

(f) Past $K_d \approx X_T$ the slowdown really does fall away. Explain why, using (a) — and say why this is not a way of escaping the load.

(g) Part (a) integrated over *all* X. But the system only ever travels from 0 to X_T, so what it actually pays on the way up is

$$\int_0^{X_T} R(X)\,dX \;=\; _____$$

Evaluate it at $K_d = 1$ and at $K_d = 100$, with $p_{\text{tot}} = 300$ and $X_T = 1000$. Thursday's sweep was *nearly* flat, not exactly flat. Say which part of that your two numbers account for — and then name one feature of that curve they do **not** account for. (Look at the shape of the sweep on the left of the figure, not just its ends.)

Before Tuesday

Aoki *et al.*, *Nature* 570:533 (2019) — an antithetic integral controller built from two molecules that destroy each other. Read the step responses.

Then ask the question this session has prepared you for: **what does dilution do to the guarantee?**

PS8 goes out at the end of this session and is due 19 November. It covers sessions 18, 20, 21 and 22.